



Reference (5)

Jeff (CHI-HSUAN HO)

The subsequent slides only showcase a few parts of the work materials and results.



Chip Image Gridding – Signal Intensity Extraction, Collection, and Correction Software

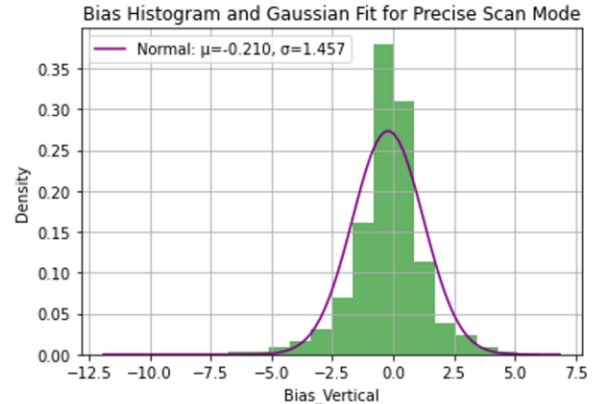
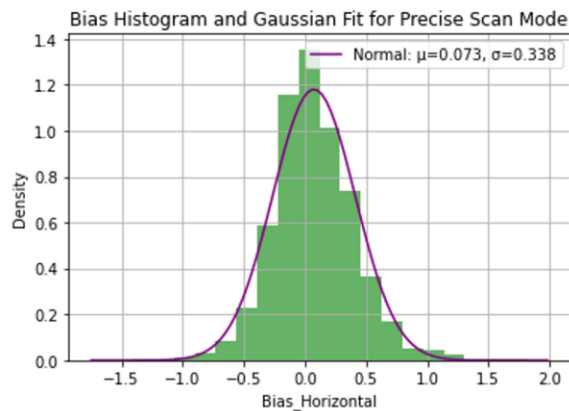
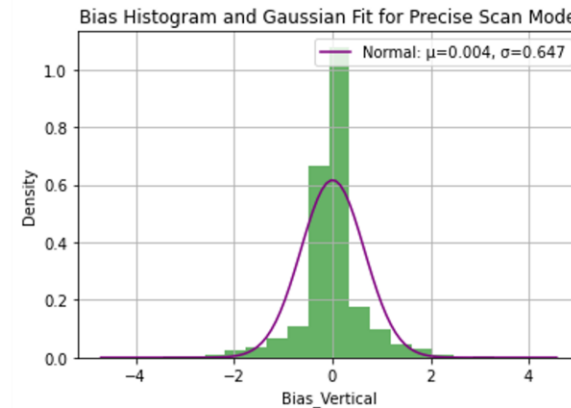
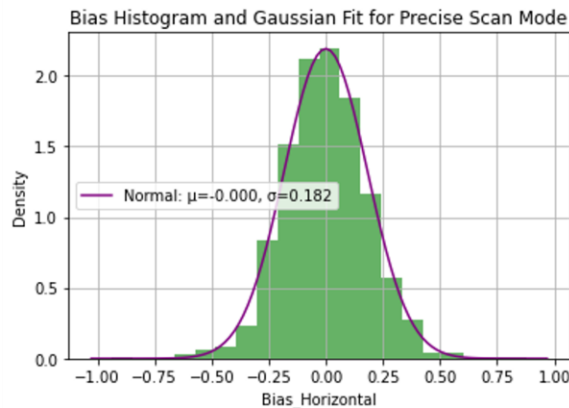
Jeff (CHI-HSUAN HO)

- Intensity Extraction Techniques Development

- Parameter collection for different scan mode.

Sample: 5 YZ01 chips (5x5 FOVs) x 10 runs => 1250 FOVs

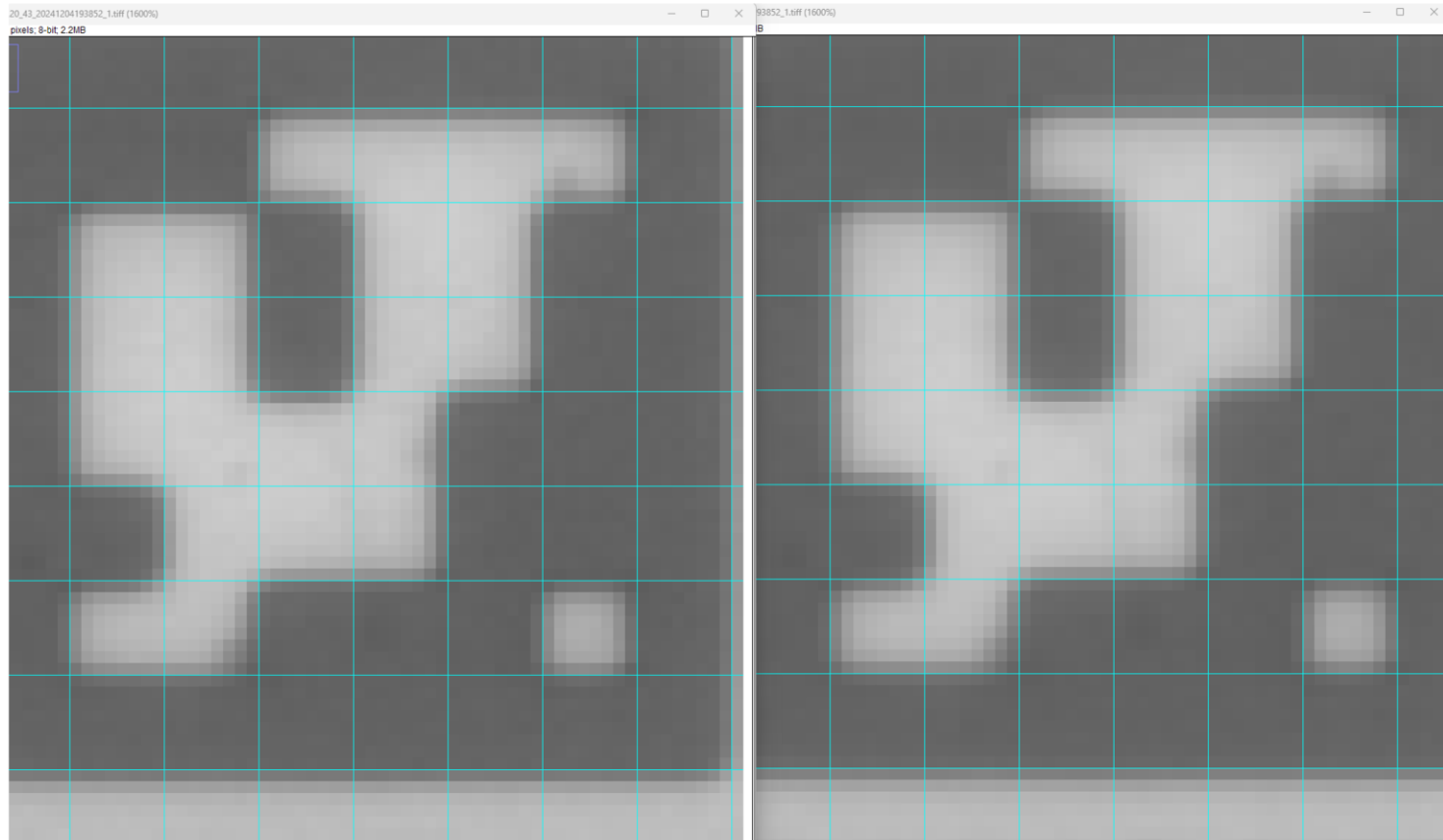
Unexpected vague between filter change.



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- Intensity Extraction Techniques Development
 - Marker recognition algo. improvements. => New algo. emphasizes symmetry more.



Old algo.

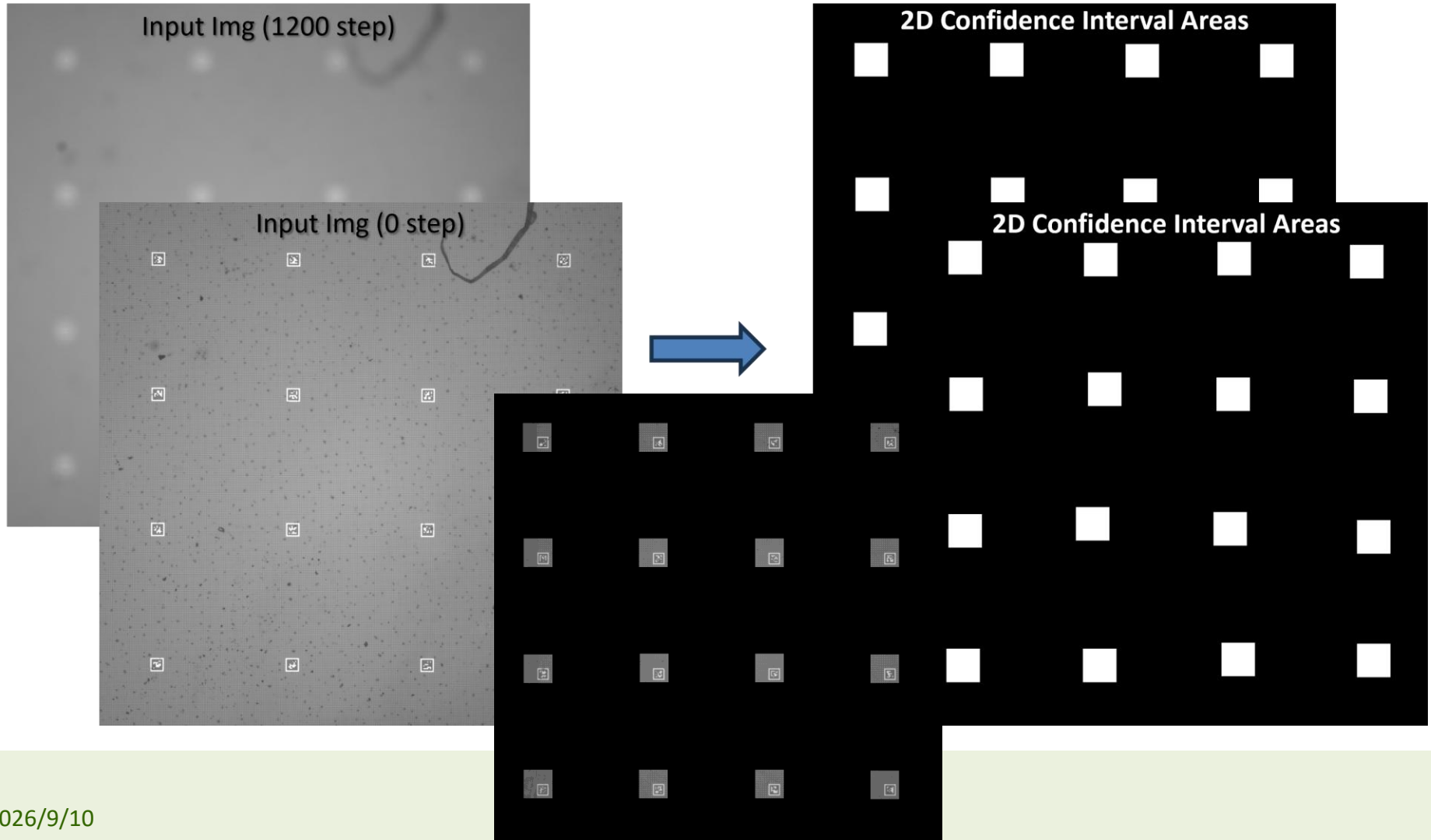
New algo.

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Chip Image Gridding

- Marker Positioning Algorithm Development



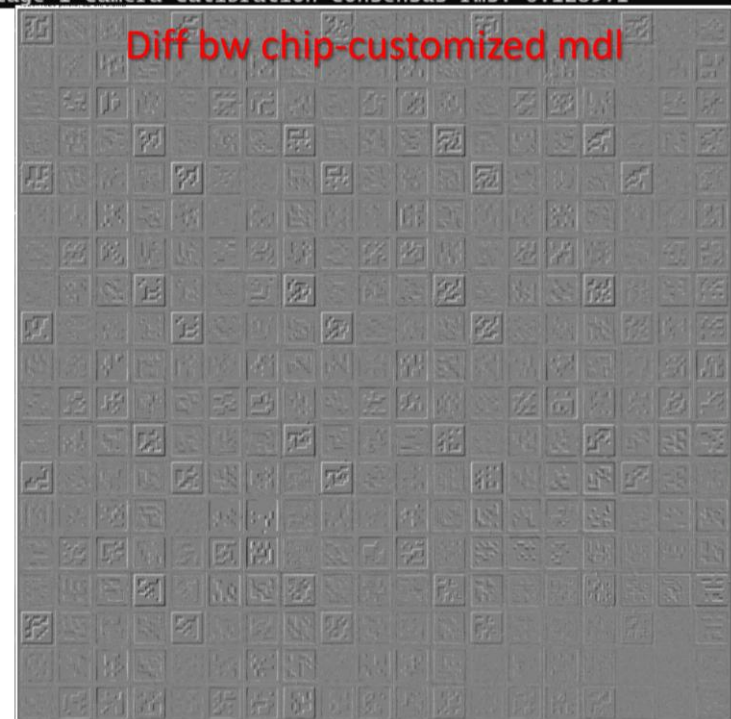
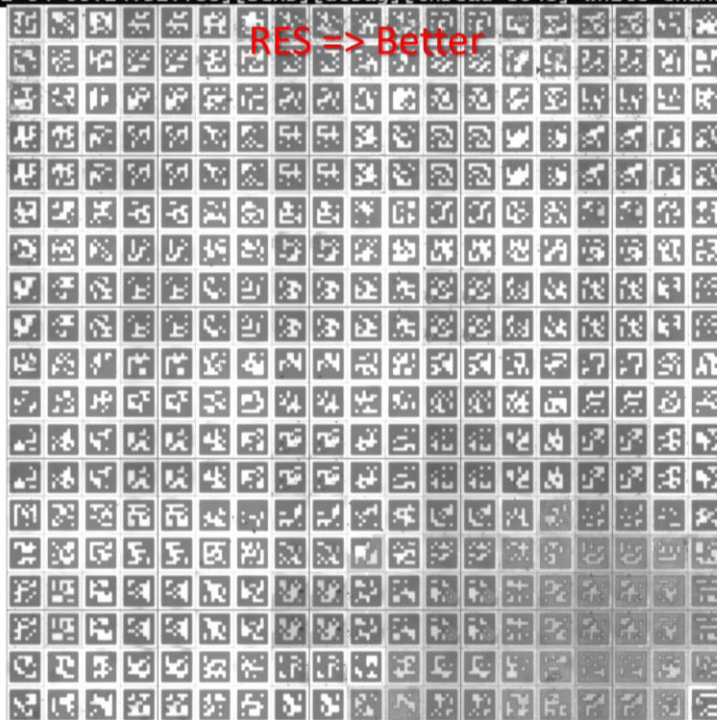
Chip Image Gridding



- Optical Calibration Techniques Evaluation by Sampling

```
[2024-12-03 15:39:07.617][SGRD][debug][thread 8048] white channel, fov id:(24, 24) end second stage marker finding process  
camera matrix:  
[90308.33478976651, 0, 2083.5;  
0, 90308.33478976651, 2083.5;  
0, 0, 1]  
distort coefficients:  
[4.405901250256348, 66.4830198744922, -0.005953858235753136, -0.0009254212662531752, -0.2134072403288333]  
[2024-12-04 00:24:32.785][SGRD][debug][thread 8048] white channel stage 1 camera calibration consensus rms: 0.128971
```

625 imgs (16 marker pos) consensus mdl => Better



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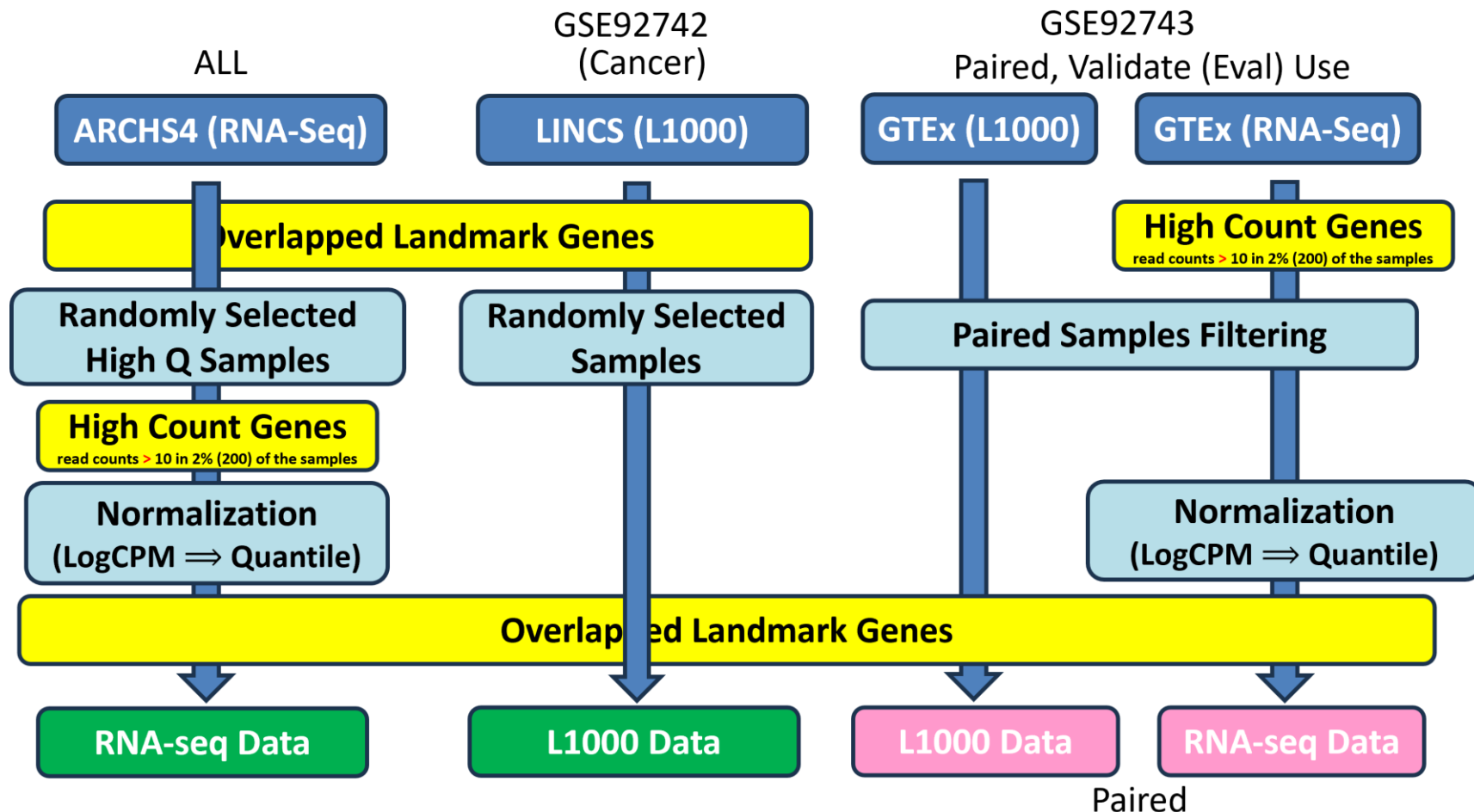
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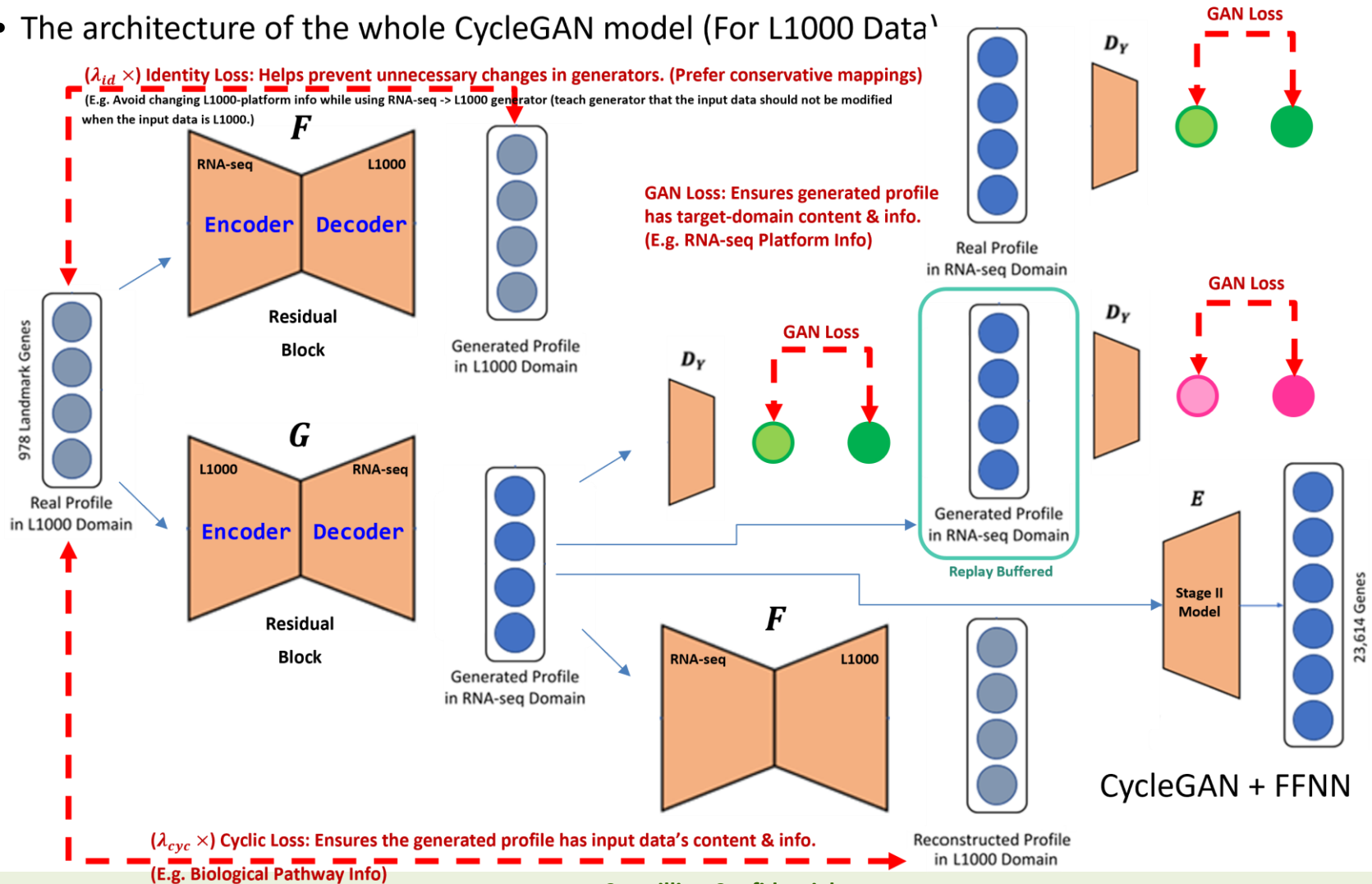
PerturbVerse – Perturbation-Conditioned Profiles and Epigenome Prediction Evaluation

Jeff (CHI-HSUAN HO)

- Profile Data Preprocessing



- The architecture of the whole CycleGAN model (For L1000 Data¹)



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- Model Structure

$$\mathcal{L}_{GAN}(G, D_Y, X, Y) = E_{y \sim p_{data}(y)} [D(y)]^2 + E_{x \sim p_{data}(x)} [1 - D(G(x))]^2$$

$$\mathcal{L}_{GAN}(F, D_X, Y, X)$$

$$\mathcal{L}_{cyc}(G, F) = E_{x \sim p_{data}(X)} [F(G(x)) - x]^2 + E_{y \sim p_{data}(Y)} [G(F(y)) - y]^2$$

To convert transformed L1000 data back into the original L1000 and vice versa.

$$\mathcal{L}_{identity}(G, F) = E_{x \sim p_{data}(X)} [F(x) - x]^2 + E_{y \sim p_{data}(Y)} [G(y) - y]^2$$

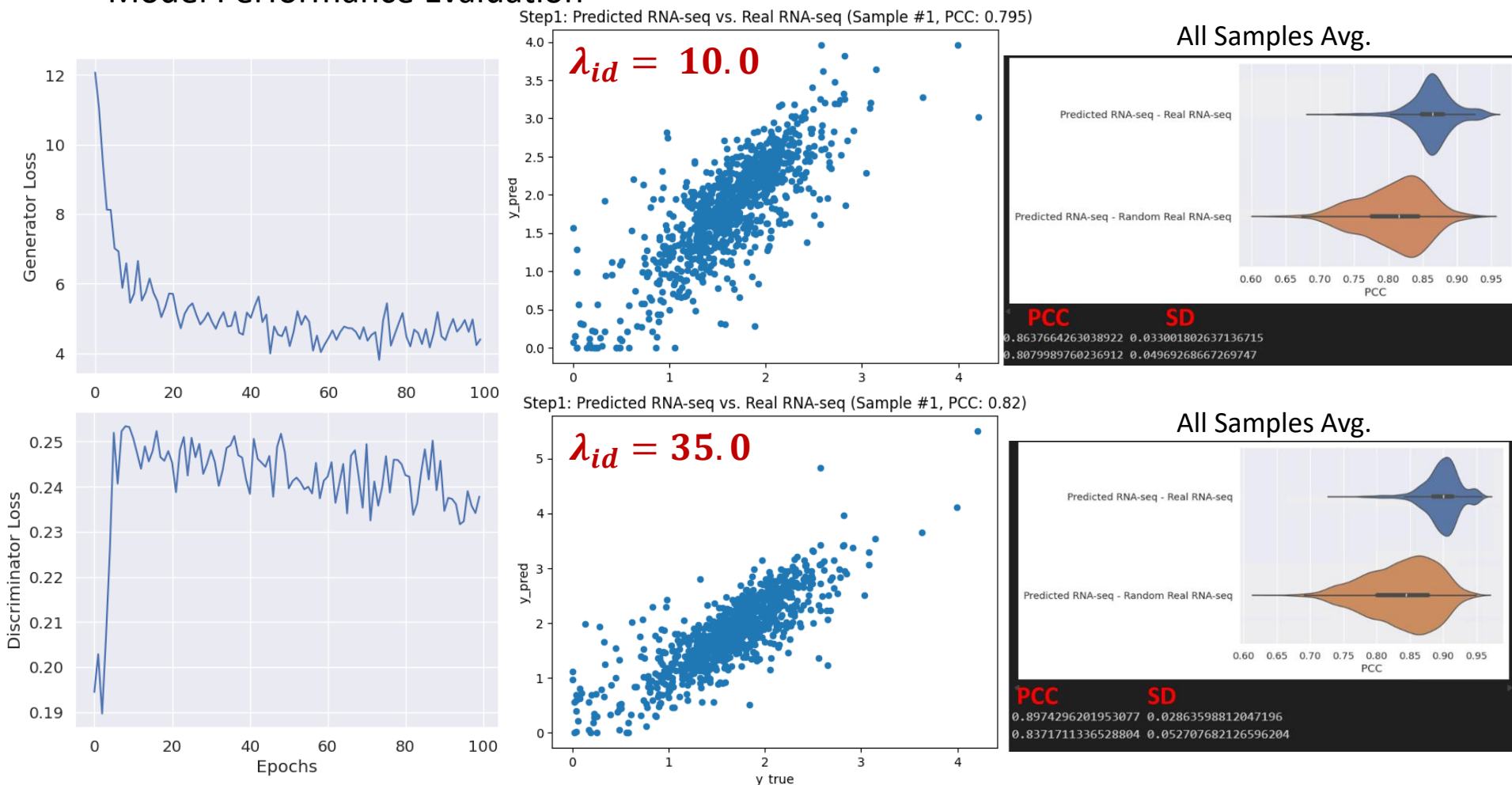
To preserve the common gene expression patterns in profiles regardless of assay platform.

$$\mathcal{L}(G, F, D_X, D_Y) = L_{GAN}(G, D_Y, X, Y) + L_{GAN}(F, D_X, Y, X) + \lambda_{cyc} * L_{cyc}(G, F) + \lambda_{identity} * L_{identity}(G, F)$$

λ_{cyc} and $\lambda_{identity}$ give relative importance to the cycle consistency loss and the identity loss.
Here, λ_{cyc} and $\lambda_{identity}$ are set to 10.

$$\min_{G, F} \max_{D_X, D_Y} \mathcal{L}(G, F, D_X, D_Y)$$

- Model Performance Evaluation



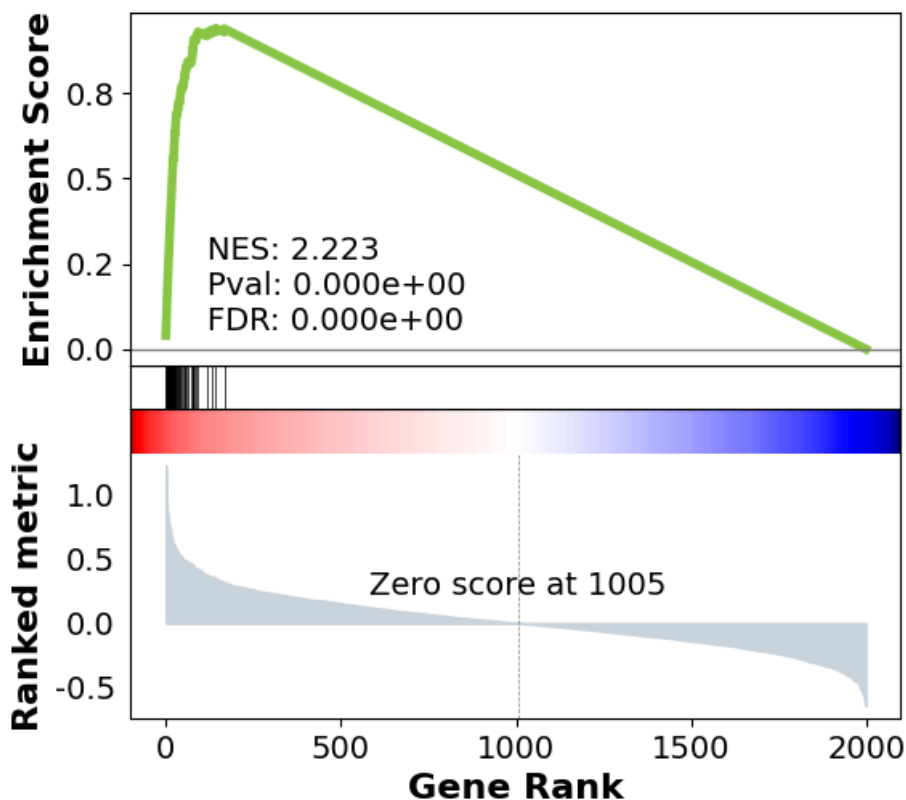
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- Further Evaluation on Simulation Data

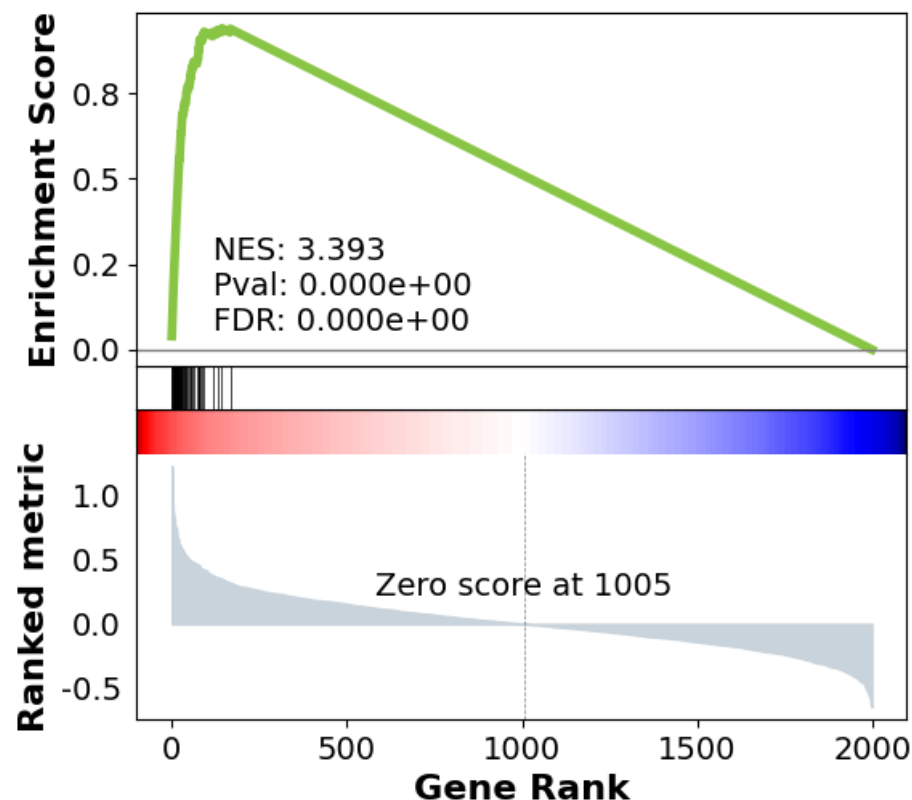
Phenotype permutation (Self-contained)

target

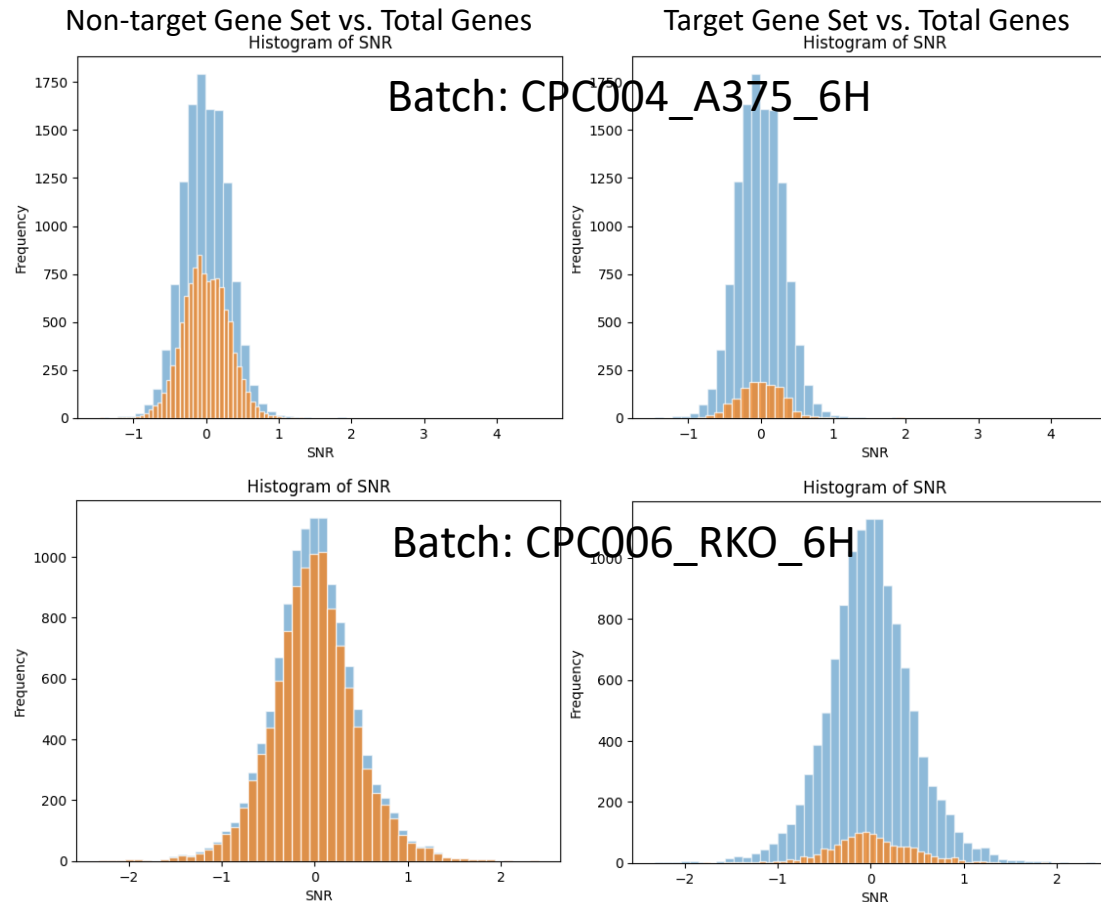


Gene set permutation (Competitive)

target



- Real Data Performance and Batch Evaluation



LINCS (L1000)

Sample Size: 32 (Cases: 3),
Total Gene Size: 11833,
Target Gene Set Size: 1263,
Non-target Gene Set Size: 10570

LINCS (L1000)

Sample Size: 20 (Cases: 2),
Total Gene Size: 11833,
Target Gene Set Size: 1263,
Non-target Gene Set Size: 10570

- Real Data Performance and Batch Evaluation

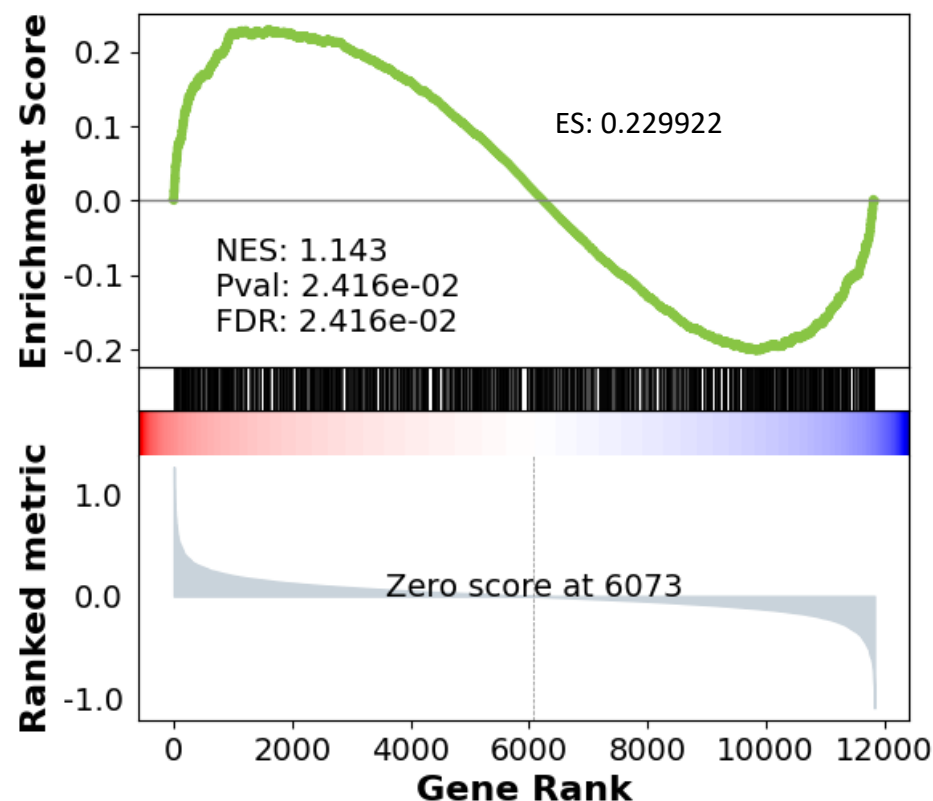
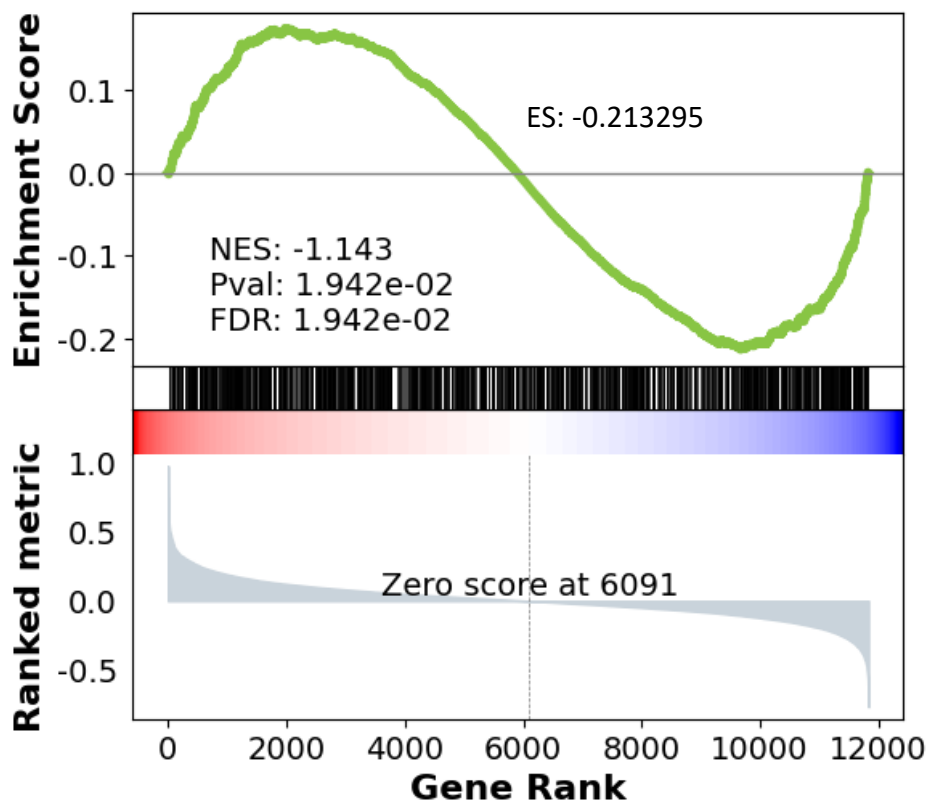
Batch: CPC004_A375_6H

NR3C1

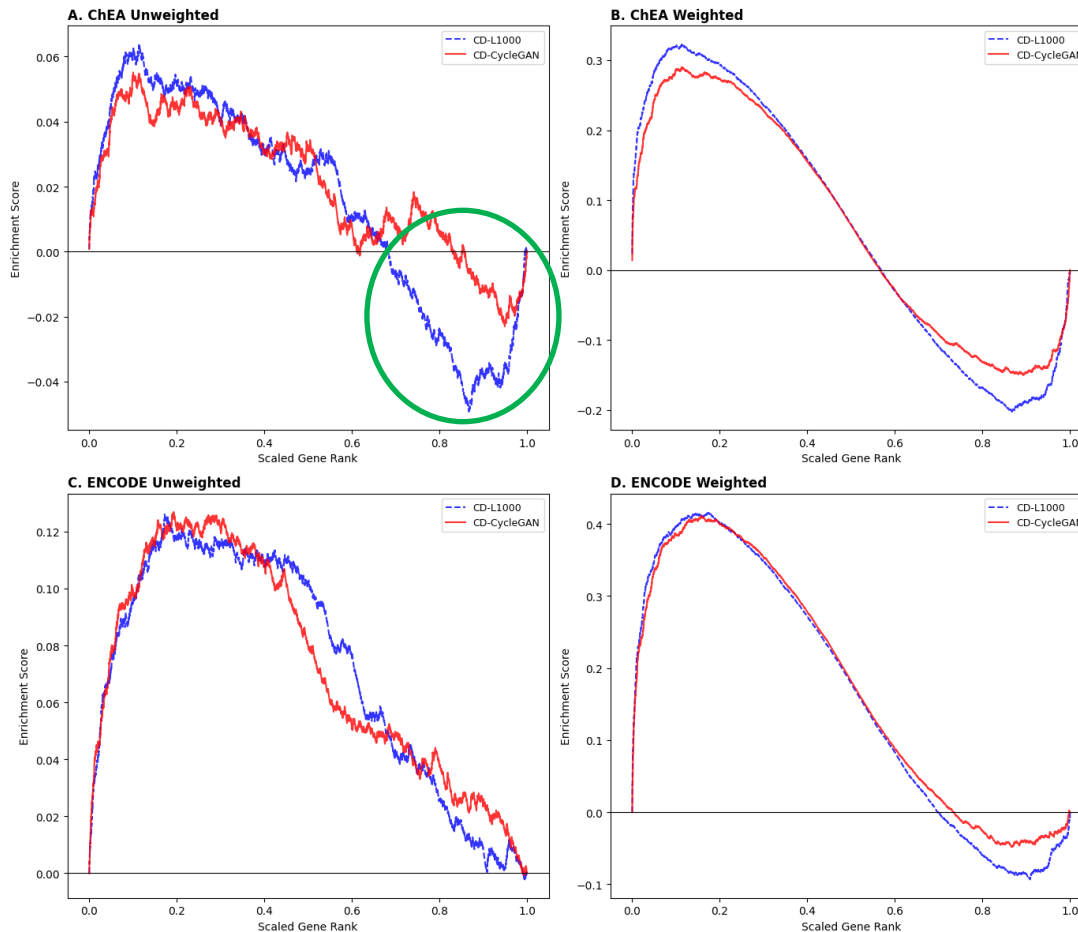
Gene Set Permutation (competitive)

NR3C1

Batch: CPC006_RKO_6H



- Real Data Performance and Batch Evaluation



All Batches Average

Control: Only ctl_vehicle (DMSO)

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